

SAI SUMANTH MUVVA

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Open to Relocation | 3+ years DE experience | [linkedin.com/in/sumanth-muvva](https://www.linkedin.com/in/sumanth-muvva) | github.com/sumanthmuvva

EDUCATION

Master of Science in Computer Science, (Indiana University Bloomington, USA) - GPA (3.8/4.0)

Aug 2022 - May 2024

Bachelor of Technology in Computer Science (KL University, India) - CGPA (9.02/10.0)

Aug 2015 - May 2019

WORK EXPERIENCE

Data Engineer-II

Aug 2021 - Jul 2022

UnitedHealth Group

- Migrated **ETLT/ETL pipelines** handling 15 TB (each insurance use case) from Teradata to **Azure Synapse Analytics**, utilizing **Azure Data Factory**.
- Innovated a **Reconciliation Framework** with **Presto** and **PySpark** to reconcile large PROD and HADR tables, saving 15 weekly hours in JIRA board.
- Coded **PySpark** on **Databricks** with Apache Spark MLlib to perform distributed data processing tasks across massive datasets. Achieved a 35% increase in the speed of data ingestion and processing.
- Applied **SCD (Type 1 & 2)** and **CDC** techniques, improving incremental data accuracy within critical financial data pipelines.
- Onboarded 6 junior data engineers and spearheaded the development of data acquisition and data integration layers for 3 key financial batches.
- Revamped **ETL workflows** within the **Enterprise Data Warehouse (EDW)** by migrating **400+ IBM DataStage jobs** from IBM scheduler to **Airflow**, leveraging **CI/CD** in **Azure DevOps**, and achieving a 30% boost in deployment speed.

Data Engineer-I

Jul 2019 - Aug 2021

UnitedHealth Group

- Developed and **fine-tuned 200+ T-SQL** queries and proposed strategic indexing and partitioning in **Teradata** leading to a 40% improvement in data access speeds in Unified Data Warehouse (UDW).
- Built complex ELT (ETLT) logic with SQL** at the staging layer in 7 projects, to calculate financial amounts in the fact table by joining 5+ normalized (3NF) dimensional tables.
- Implemented cutting-edge, raw, and business **data vault** methodologies optimizing **batch processing** times by 30%.
- Devised **60+ Power BI dashboards** utilizing DAX, enabling real-time monitoring of critical metrics such as downstream statuses, ETL flow, batch times, and issues with efficient data modeling techniques.
- Streamlined data flows for 4 Business Intelligence (BI) applications into data marts, using **Microsoft SSIS** and **MS SQL Server 2019**, to enhance data driven decision-making.
- Collaborated with a 27-member cross-functional team, including managers, product owners, BI and data analysts, to understand requirements in multiple ETL projects and deliver solutions with high business value using **Agile** methodology.

Data Analyst Intern

Jan 2019 - Jun 2019

NprodaX Technologies

- Optimized **data models** and developed custom **APIs** for a web app, ensuring **efficient data storage** and retrieval, and crafted a **Power BI** dashboard with **over 10 charts** for real-time cryptocurrency analytics display.
- Integrated and configured **Apache Kafka** streams to ingest live data feeds into Power BI, ensuring up-to-date visualizations for 150 cryptocurrencies.

TECHNICAL SKILLS

- ETL & Data Warehousing** : ETL/ELT/ETLT Pipelines, IBM DataStage (Designer, Director), Teradata, EDW, SSIS, Apache Airflow, Databricks, Kafka.
- Programming & Databases** : Python, SQL/T-SQL, Presto, PySpark, Java, C, UNIX, Bash, MySQL, PostgreSQL, NoSQL (DynamoDB, Cosmos DB).
- Visualization & Reporting** : Microsoft Power BI, Tableau, Grafana.
- Cloud Platforms** : AWS (S3, Lambda, Glue, EMR, RDS, Redshift, Athena, CDK), Azure (Data Factory, Synapse), Snowflake.
- DevOps & Other** : Git, Docker, Jenkins, CI/CD pipelines, Terraform, Azure DevOps, SDLC (Agile).

CERTIFICATIONS

- AWS Certified - Solutions Architect Associate (SAA-C03)**.
- Microsoft Certification - Azure Data Engineer Associate (DP-203)** – (Expected Apr 10, 2024).

PROJECTS

YouTube Insights Analysis and Visualization (ETL) [AWS - EC2, S3, IAM, Glue, Lambda, Kinesis, Athena, QuickSight].

Jun 2023 - Aug 2023

- Engineered an end-to-end scalable ETL data pipeline using **AWS** cloud infrastructure and Kaggle data, that help clients to get enriched YouTube **user insights** and engagement analytics for targeted advertising strategies.
- Implemented AWS **Lambda** and **Athena** for efficient, serverless data processing, integrated with **S3** and **Glue**, culminating in a **QuickSight** dashboard with 6+ key metrics empowering data-driven decision-making.

Home Credit Default Risk Analysis [Python, Sklearn, MLP, Feature Selection, Hyperparameter Tuning, TensorFlow].

Jan 2023 - Apr 2023

- Architected an Enhanced Multi-Layer Perceptron (MLP), achieving 68.06% accuracy with gradient descent, feature selection and hyperparameter tuning via grid search boosting F1 score to 0.7029.
- Leveraged Deep Learning and ensemble methods, enhancing model performance to a public score of 0.7537, refining lending strategies.

LEADERSHIP EXPERIENCE

- Technical Events Lead**, SAMYAK-19 Technical Festival at KL University, led a team of 12 event organizers in hosting tech challenges.
- Mentor**, ServiceNow Developer Certification (SNAPP), guided 240 students through the certification process at KL University.